

SPOTIFY TRACKS DATASET ANALYSIS USING SQL

EASY QUERIES

Question 1: What are the top 10 most popular tracks in the dataset?

```
SELECT track_name, artists, popularity
FROM spotify_project.spotify_tracks
ORDER BY popularity DESC
LIMIT 10;
```



track_name	artists	popularity
Unholy (feat. Kim Petras)	Sam Smith;Kim Petras	100
Unholy (feat. Kim Petras)	Sam Smith;Kim Petras	100
Quevedo: Bzrp Music Sessions, Vol. 52	Bizarrap;Quevedo	99
I'm Good (Blue)	David Guetta;Bebe Rexha	98
I'm Good (Blue)	David Guetta;Bebe Rexha	98
La Bachata	Manuel Turizo	98
La Bachata	Manuel Turizo	98
I'm Good (Blue)	David Guetta;Bebe Rexha	98
La Bachata	Manuel Turizo	98
La Bachata	Manuel Turizo	98

Observation: The data shows that "Unholy" by Sam Smith and Kim Petras holds the absolute top spot with a maximum popularity score of 100.

Question 2: What are the top 10 most popular tracks that do not contain explicit content?

```
SELECT track_name, artists, popularity, explicit
FROM spotify_project.spotify_tracks
WHERE explicit = 'False'
ORDER BY popularity DESC
LIMIT 10;
```

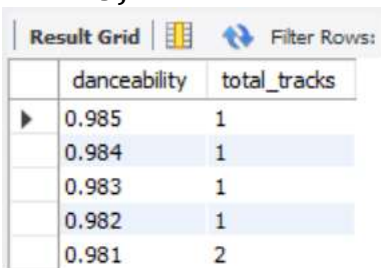


track_name	artists	popularity	explicit
Unholy (feat. Kim Petras)	Sam Smith;Kim Petras	100	False
Unholy (feat. Kim Petras)	Sam Smith;Kim Petras	100	False
Quevedo: Bzrp Music Sessions, Vol. 52	Bizarrap;Quevedo	99	False
La Bachata	Manuel Turizo	98	False
La Bachata	Manuel Turizo	98	False
La Bachata	Manuel Turizo	98	False
La Bachata	Manuel Turizo	98	False
Tití Me Preguntó	Bad Bunny	97	False
Tití Me Preguntó	Bad Bunny	97	False
Tití Me Preguntó	Bad Bunny	97	False

Observation: By filtering the dataset to show only non-explicit tracks, we see that songs like "I'm Good (Blue)" are removed from the rankings. This allows clean global hits to move up into the top tier of the popularity list.

Question 3: How many total tracks are there for each unique danceability score?

```
SELECT danceability, COUNT(*) AS total_tracks
FROM spotify_project.spotify_tracks
GROUP BY danceability
ORDER BY danceability DESC
LIMIT 5;
```



danceability	total_tracks
0.985	1
0.984	1
0.983	1
0.982	1
0.981	2

Observation: The highest danceability score in the dataset is 0.985, and only one song has it. This shows that tracks with a near-perfect dance score are extremely rare.

Question 4: What are the top 5 longest tracks in the dataset?

```
SELECT track_name, artists, duration_ms
FROM spotify_project.spotify_tracks
ORDER BY duration_ms DESC
LIMIT 5;
```

	track_name	artists	duration_ms
▶	Unity (Voyage Mix) Pt. 1	Tale Of Us	5237295
	Crossing Wires 002 - Continuous DJ Mix	Timo Maas	4789026
	Crossing Wires 002 - Continuous DJ Mix	Timo Maas	4789026
	The Lab 03 - Continuous DJ Mix Part 1	Seth Troxler	4730302
	Amnesia Ibiza Underground 10 DJ Mix	Loco Dice	4563897

Observation: The longest tracks in the dataset are all long DJ mixes, with the longest one being "Unity (Voyage Mix) Pt. 1" at around 87 minutes.

Question 5: Which songs by The Weeknd belong specifically to the 'pop' genre?

```
SELECT DISTINCT track_name, artists, track_genre
FROM spotify_project.spotify_tracks
WHERE track_genre = 'pop' AND artists = 'The Weeknd';
```

	track_name	artists	track_genre
▶	Blinding Lights	The Weeknd	pop
	The Hills	The Weeknd	pop
	After Hours	The Weeknd	pop
	Out of Time	The Weeknd	pop
	Take My Breath - Single Version	The Weeknd	pop
	Call Out My Name	The Weeknd	pop
	Save Your Tears	The Weeknd	pop
	Die For You	The Weeknd	pop

Observation: Using the DISTINCT keyword successfully removes repetitive entries caused by duplicate album releases. The filtered list highlights major pop hits by The Weeknd, such as "Blinding Lights" and "Die For You".

Question 6: Which tracks are strictly instrumental (tracks containing no vocals at all)?

```
SELECT DISTINCT track_name, artists, instrumentalness
FROM spotify_project.spotify_tracks
WHERE instrumentalness > 0.5;
```

	track_name	artists	instrumentalness
▶	And I Love Her	Kurt Cobain	0.833
	Chi Mai	Joseph Sullinger	0.961
	Mujer con Abanico	Agustín Amigó;Nylonwings	0.905
	A Thousand Years	Eddy Tyler	0.543
	Love Nwantiti	Andrew Foy	0.886
	Ramparts	John Frusciante	0.882
	Tamacun	Rodrigo y Gabriela	0.856
	Fake a smile	Wayne Mack	0.671
	River Flows In You	Sungha Jung	0.801
	Guaranteed - Humming Version	Eddie Vedder	0.879

Observation: Filtering by an instrumentalness score greater than 0.5 successfully pulls a large number of vocal-free tracks.

Question 7: Which songs begin with the word "Love"?

```
SELECT DISTINCT track_name, artists
FROM spotify_project.spotify_tracks
```

WHERE track_name LIKE 'Love%';

track_name	artists
Love Like This	Ben Rector
Love Song	Sara Bareilles
Love Story	Tyler Ward;Karis;Ray Lorraine
Love Someone	Jason Mraz
Love You To Death	Chord Overstreet
Love Nwantiti	Andrew Foy
Love	Matt White
Love Me Like You Do	Boyce Avenue
Love	Greyboy;Nino Moschella
Love Like This - 45 Edit	Bacao Rhythm & Steel Band

Observation: Using the LIKE operator with a filter allows us to see that around 400 unique tracks in the dataset begin exactly with the word "Love".

Question 8: Which unique tracks belong to either the 'rock' or 'indie' genres?

```
SELECT DISTINCT track_name, artists, track_genre
FROM spotify_project.spotify_tracks
WHERE track_genre IN ('rock', 'indie');
```

track_name	artists	track_genre
Aisay Kaisay	Hasan Raheem;Abdullah Kasumbi	indie
Manike (From "Thank God")	Yohani;Jubin Nautiyal;Tanishk Bagchi;Rashmi Vir...	indie
Gul	Anuv Jain	indie
I Wanna Be Yours	Arctic Monkeys	indie
Apocalypse	Cigarettes After Sex	indie
Aaj Na	Ritviz	indie
Bikhra	Abdul Hannan;Rovalio	indie
Nasha (Equals Sessions)	Faridkot;Amar Jalal	indie
Roz Roz	The Yellow Diary;Shilpa Rao	indie
Formula	Labrinth	indie

Observation: Using the IN operator allows for clean filtering across multiple text categories simultaneously, pulling a combined list of rock and indie tracks.

MEDIUM QUERIES

Question 1: What are the top 5 most popular music genres based on average popularity?

```
SELECT track_genre, AVG(popularity) AS average_popularity
FROM spotify_project.spotify_tracks
GROUP BY track_genre
ORDER BY average_popularity DESC LIMIT 5;
```

track_genre	average_popularity
pop-film	59.2830
k-pop	56.8960
chill	53.6510
sad	52.3790
grunge	49.5940

Observation: The results show that pop-film and k-pop lead with the highest average popularity, while genres like chill and sad follow closely behind.

Question 2: What is the popularity range for active tracks (popularity greater than 0) across different music genres?

```
SELECT track_genre, MAX(popularity) AS max_pop, MIN(popularity) AS min_pop,
       ROUND(MAX(popularity) - MIN(popularity), 2) AS popularity_range
FROM spotify_project.spotify_tracks
WHERE popularity > 0
GROUP BY track_genre
```

LIMIT 10;

	track_genre	max_pop	min_pop	popularity_range
▶	acoustic	82	1	81
	afrobeat	75	3	72
	alt-rock	93	1	92
	alternative	93	1	92
	ambient	84	1	83
	anime	83	1	82
	black-metal	58	8	50
	bluegrass	69	12	57
	blues	84	1	83
	brazil	82	1	81

Observation: The results show that genres like alt-rock and alternative have a very wide popularity range of 92 points. This indicates a massive gap between their top viral hits and their least popular active tracks.

Question 3: What are the top 5 music genres with the longest average track duration in minutes?

```
SELECT track_genre,
        ROUND(AVG(duration_ms) / 60000, 2) AS avg_duration_minutes
FROM spotify_project.spotify_tracks
GROUP BY track_genre
ORDER BY avg_duration_minutes DESC LIMIT 5;
```

	track_genre	avg_duration_minutes
▶	detroit-techno	6.20
	minimal-techno	6.15
	chicago-house	6.11
	breakbeat	5.36
	iranian	5.33

Observation: Electronic genres like detroit-techno and minimal-techno have the longest tracks, averaging over 6 minutes.

Question 4: Which top 8 artists have the highest count of high-energy, danceable tracks?

```
SELECT artists, COUNT(*) AS high_energy_dance_tracks
FROM spotify_project.spotify_tracks
WHERE danceability > 0.8 AND energy > 0.8
GROUP BY artists
ORDER BY high_energy_dance_tracks DESC
LIMIT 8;
```

	artists	high_energy_dance_tracks
▶	Daddy Yankee;Bad Bunny	58
	J Balvin;Maria Becerra	51
	Anuel AA;KAROL G	44
	Daddy Yankee	29
	J Balvin	19
	Paul Johnson	16
	Galantis	15
	Don Omar	15

Observation: Artists like Daddy Yankee, Bad Bunny, and J Balvin dominate the list. This shows that these specific artists consistently produce the highest volume of heavily danceable, high-energy music in the entire dataset.

Question 5: Explicit or non-explicit tracks, which have a higher average popularity score?

```
SELECT explicit,
        ROUND(AVG(popularity), 2) AS avg_popularity
FROM spotify_project.spotify_tracks
```

GROUP BY explicit;

	explicit	avg_popularity
▶	False	32.94
	True	36.45

Observation: Explicit tracks have a noticeably higher average popularity score (36.45) compared to non-explicit tracks (32.94). This indicates that songs with explicit content perform slightly better on average within this dataset.

Question 6: Which top 8 artists have tracks spanning across the highest number of unique genres?

```
SELECT artists, COUNT(DISTINCT track_genre) AS unique_genre_count
FROM spotify_project.spotify_tracks
GROUP BY artists
ORDER BY unique_genre_count DESC
LIMIT 8;
```

	artists	unique_genre_count
▶	Badfinger	9
	Derek & The Dominos	8
	Fear, and Loathing in Las Vegas	7
	Allman Brothers Band	7
	Anupam Roy	7
	Prateek Kuhad	7
	Peter Dinklage	7
	Ritviz	7

Observation: Badfinger (9 genres) and Derek & The Dominos (8 genres) lead the dataset in unique genre songs. They are closely followed by artists like Anupam Roy, Prateek Kuhad, and Ritviz, who all span across 7 unique genres.

HARD QUERIES

Question 1: Which tracks have a popularity score higher than the average popularity of all tracks in the entire dataset?

```
SELECT track_name, artists, popularity
FROM spotify_project.spotify_tracks
WHERE popularity > (
    SELECT AVG(popularity)
    FROM spotify_project.spotify_tracks)
ORDER BY popularity DESC LIMIT 5;
```

	track_name	artists	popularity
▶	Unholy (feat. Kim Petras)	Sam Smith;Kim Petras	100
	Unholy (feat. Kim Petras)	Sam Smith;Kim Petras	100
	Quevedo: Bzrp Music Sessions, Vol. 52	Bizarrap;Quevedo	99
	I'm Good (Blue)	David Guetta;Bebe Rexha	98
	I'm Good (Blue)	David Guetta;Bebe Rexha	98

Observation: Top tracks like "Unholy" and "Quevedo" easily beat the average popularity of the dataset, showing they are among the platform's biggest hits.

Question 2: How many high-danceability tracks versus low-danceability tracks does each artist have?

```
SELECT artists, COUNT(*) AS total_tracks,
    SUM(CASE WHEN danceability > 0.7 THEN 1 ELSE 0 END) AS high_danceability_count,
    SUM(CASE WHEN danceability <= 0.7 THEN 1 ELSE 0 END) AS low_danceability_count
FROM spotify_project.spotify_tracks
GROUP BY artists
```

ORDER BY high_danceability_count DESC;

	artists	total_tracks	high_danceability_count	low_danceability_count
►	Feid	202	179	23
	Don Omar	137	132	5
	Burna Boy	116	110	6
	CoComelon	122	101	21
	Wisin & Yandel	87	84	3
	Daddy Yankee	154	83	71
	J Balvin	104	82	22
	Boris Brejcha	83	81	2

Observation: Artists like Feid and Don Omar dominate the high-danceability category, with the vast majority of their tracks scoring above 0.7.

Question 3: For each genre, what is the difference between the highest popularity score and the average popularity score?

```
SELECT track_genre,
       MAX(popularity) AS max_popularity,
       ROUND(AVG(popularity), 2) AS avg_popularity,
       ROUND(MAX(popularity) - AVG(popularity), 2) AS popularity_difference
FROM spotify_project.spotify_tracks
GROUP BY track_genre
ORDER BY popularity_difference DESC
LIMIT 10;
```

	track_genre	max_popularity	avg_popularity	popularity_difference
►	latin	98	8.30	89.70
	reggae	98	20.63	77.37
	dance	100	22.69	77.31
	rock	96	19.00	77.00
	reggaeton	98	23.86	74.14
	latino	98	25.66	72.34
	country	89	17.03	71.97

Observation: Genres like latin and reggae show the largest popularity gaps.

Question 4: Which artists have an average danceability higher than the overall average danceability of all tracks?

```
SELECT artists,
       ROUND(AVG(danceability), 2) AS artist_avg_danceability
FROM spotify_project.spotify_tracks
GROUP BY artists
HAVING AVG(danceability) > (
    SELECT AVG(danceability)
    FROM spotify_project.spotify_tracks)
ORDER BY artist_avg_danceability DESC
LIMIT 10;
```

	artists	artist_avg_danceability
►	Casper & Col'ta	0.98
	Vybz Kartel;Tommy Lee	0.98
	DJ Pierre;My Digital Enemy;Hatiras	0.98
	BaianaSystem;Tropkillaz	0.97
	Moliy;Juls	0.97
	Bigg Robb;Mz Jackson	0.97
	Cajmere;Claude VonStroke	0.97

Observation: "This query identifies artists whose average danceability is higher than the overall average danceability of the dataset.". Artists like Casper & Co'ta and Vybz Kartel achieve near-maximum average danceability scores (0.98).